

1. Title of the Invention

Microcontroller-Free Wireless control for Home & Industrial Automation.

2. Field of Invention

The present invention relates to the field of **home and industrial automation**, more particularly to wireless control systems that operate **without microcontrollers**, instead utilizing fixed-address encoder/decoder integrated circuits (ICs), demultiplexers, and silicon-controlled rectifier (SCR) latch circuits for reliable and long-range operation using **433 MHz RF transmission**.

3. Background of the Invention

Conventional home automation systems predominantly employ programmable microcontrollers (e.g., Arduino, ESP8266) with RF or Wi-Fi modules. These systems, while functional, introduce complexity due to firmware development, require consistent internet connectivity, and are costlier to implement. Additionally, smartphone dependency and setup intricacies render such systems unsuitable for elderly or non-technical users.

Infrared (IR)-based remote control systems suffer from line-of-sight limitations and short operating range (typically under 10 meters). ZigBee and Wi-Fi-based solutions offer longer range but at higher cost, network dependency, and complex configuration. Furthermore, these systems are power-intensive and fail to retain the device's state during power loss unless expensive memory-backed components are used.

Some prior designs use fixed-code encoder/decoder pairs (e.g., HT12E/HT12D) for remote operation but lack **latching capability** or master-reset functionality. No widely adopted or patented architecture has successfully implemented a complete microcontroller-free, RF-based solution with **SCR latches** that **retain device state**, support **master-off shutdown**, and offer **scalability up to 16 outputs**.

4. Object of the Invention

The primary objective of the invention is to develop a **low-cost, microcontroller-free wireless automation system** that provides:

- **Long-range operation** (up to 100 meters) without line-of-sight limitations;
- **Multi-device control** (up to 16 appliances) via a 4×4 keypad;
- **State retention using SCR latches**, eliminating need for continuous power or signal;
- **Universal master-off control** to safely power-off all devices;
- **No software dependency**, enabling plug-and-play usage even for non-technical users;
- **Significant cost reduction** (approx. ₹2,900) compared to microcontroller-based systems (~₹6,700).

This invention bridges the gap between hardware simplicity, user accessibility, and industrial robustness.

5. Summary of the Invention

The invention consists of a **transmitter unit** and a **receiver unit**. A 4×4 keypad, when actuated, generates a 4-bit code corresponding to the pressed key. This binary code is passed to an HT12E encoder IC configured with a fixed 8-bit address. The encoder outputs a 12-bit serial signal, which is transmitted via a **433 MHz RF module**.

On the receiver side, an HT12D decoder IC—configured with the matching address—validates the received signal and outputs the 4-bit data to a **74HC154 demultiplexer**. The demultiplexer decodes the 4-bit input into 1-of-16 output lines, where each line drives a **NOT gate** followed by a **SCR latch** connected to a relay or load.

Each SCR latch retains its ON state until a reset condition is applied. A special “master-off” key on the keypad sends a unique code which activates a transistor circuit that disrupts gate current across all SCRs, forcing them to turn off. This allows global shutdown of all appliances with a single command.

6. Brief Description of Drawings

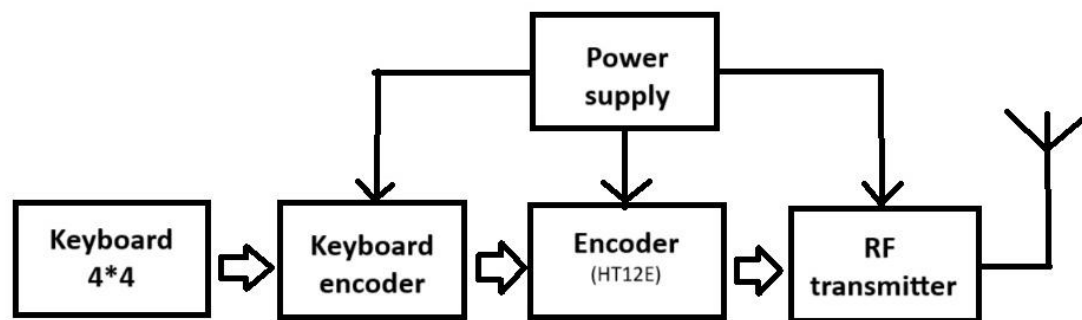


Fig. 1 -Block diagram of Transmitter

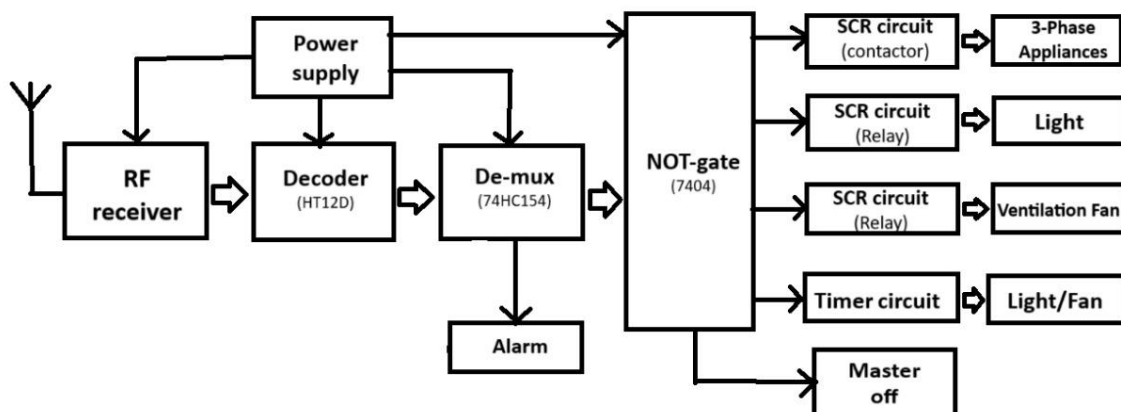


Fig 2. Block diagram of receiver

7. Detailed Description of the Invention

Transmitter Section

The user interface consists of a **4×4 membrane keypad** (16 keys), with each keypress generating a unique 4-bit binary code. This code is directly applied to the data inputs (D0–D3) of a **HT12E encoder IC**, which also contains 8 address inputs set via DIP switches or soldered logic levels.

Upon activation, the HT12E serializes the data into a 12-bit signal and transmits it via a **433 MHz RF module** using ASK modulation. The module includes a $\lambda/4$ antenna (approx. 17.3 cm) for optimized transmission. The entire unit can be powered via a 9 V battery.

Receiver Section

1. RF Reception and Decoding

- A **433 MHz RF receiver module** receives the transmitted data.
- The output is fed to an **HT12D decoder IC**, configured with a matching 8-bit address to that of the HT12E transmitter.
- Upon valid transmission, HT12D outputs a corresponding 4-bit data word (D0–D3), and activates the **VT** (Valid Transmission) pin for signaling.

2. Demultiplexer and Line Selection

- The 4-bit data is applied to a **74HC154** 4-to-16 demultiplexer.
- Exactly one of its 16 output lines (**Y0 to Y15**) goes active LOW based on the key pressed.
- These active-low lines are **inverted** using a **7404 NOT gate** (6-inverter hex package), producing an active-HIGH logic level to drive gate circuits.

3. SCR Latching Control

Each demux output drives a separate **BT169D SCR** circuit for latching operation:

- The **inverted demux output** is connected to the **gate** of the SCR.
- **Anode** of the SCR connects to one side of a **7 A relay coil**; the other side is tied to **+5 V**.
- **Cathode** is grounded.
- When the SCR gate receives current $\geq I_G \approx 200 \mu\text{A}$, it latches ON and energizes the relay.
- The relay contacts then switch the **230 VAC appliance**.
- The SCR remains latched ($I_A > I_H \approx 2\text{--}5 \text{ mA}$) until power is interrupted or **master-off** resets it.
- This configuration offers **state retention without software** or memory.

4. Master-OFF Mechanism

- The “**M_OFF**” signal (master-off) is triggered by a special demux output (**Y0**) on key "1" press.
- It drives a **S8550 PNP transistor (Q2)**, which pulls all SCR gates or anodes LOW, cutting latching current below I_H .
- This forces all SCRs OFF, instantly turning off every connected appliance.

5. Timer-Controlled Output (Auto-OFF)

- A dedicated demux output triggers a **monostable 555 timer** circuit (U13), implemented using:
 - $R4 = 33\text{ k}\Omega$, $C4 = 470\text{ }\mu\text{F} \rightarrow \sim 17\text{ s}$ ON time.
 - Q3 (BC547B) acts as a switch to energize the connected relay.
 - After timeout, the 555 output resets, turning OFF the load.
- Useful for **lamps, water pump, or exhaust fans**.

6. Fan Speed Control (Triac-Based Phase Modulation)

- Two demux outputs activate separate **phase-angle control circuits** using:
 - **Diac (DB3)** and **Triac (BTA12)** for AC control.
 - Outputs control **FAN_HIGH** and **FAN_LOW** speed states.
- Each triac output may be paired with its own **SCR** or relay for safe latching.

7. Alarm or Buzzer Output

- Another demux output triggers a **5 V buzzer** for audible alerts or system feedback.
- Can be used for **notifications, fault indication, or emergency**.

8. Additional Load Types

- Receiver can also drive **contactors** using relay.
- This enhances applicability for workshops, motor control, or plant automation.

8. Claims

9. Figures

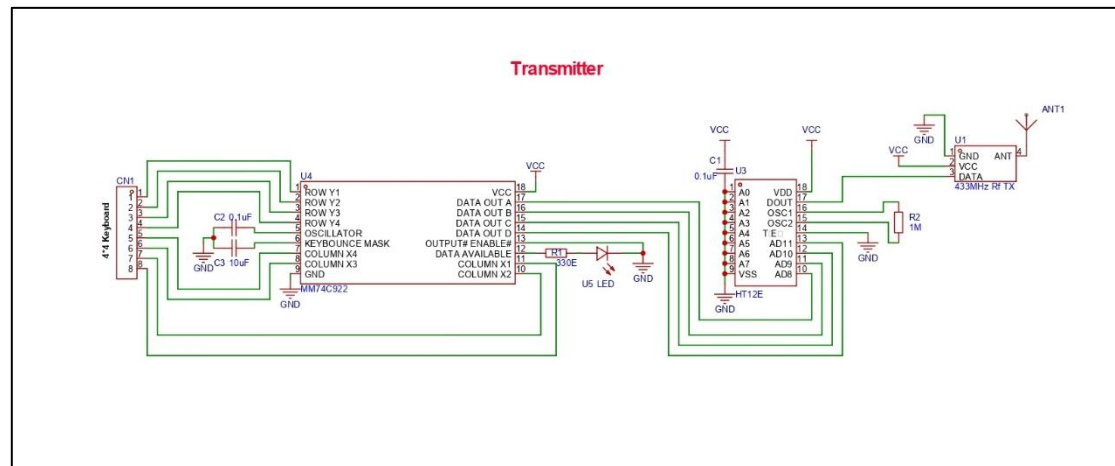


Fig. 3- Circuit diagram of Transmitter

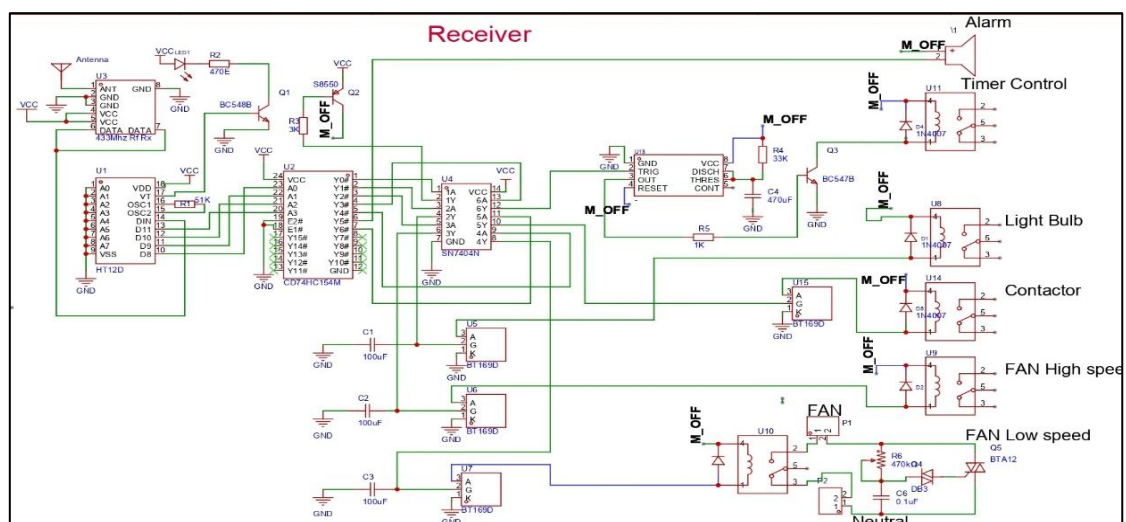


Fig. 4 – Circuit diagram of receiver

10. Abstract (≤150 words)

This invention presents a wireless automation system that operates entirely without microcontrollers, utilizing HT12E/HT12D encoder–decoder ICs, a 433 MHz RF communication link, and SCR-based latching circuits. A 4×4 keypad enables the control of up to 16 independent devices. Each keypress is encoded and transmitted wirelessly; at the receiver end, the signal is decoded and routed through a demultiplexer to trigger the corresponding SCR latch, ensuring persistent device state. A dedicated master-off key allows simultaneous shutdown of all outputs. The system achieves a wireless range of up to 100 meters and offers a cost-effective, robust, and user-friendly solution for both home and industrial automation applications.

11. Applications

- **Home Use:** Control lights, fans, water pumps, EV chargers, and security systems.
- **Industrial Use:** Activate machines, conveyor belts, HVACs, and alarm systems via contactors.
- **Agricultural Use:** Wirelessly operate irrigation motors and valves.
- **Educational Use:** Demonstrate digital logic, RF communication, and SCR-based latching.